

COMPLEXITY AND RECURRENCES

Algorithmic Performance and Mathematical Analysis

Author: Eng. Carlos Andrés Sierra, M.Sc.
cavirguezs@udistrital.edu.co

Full-time Adjunct Professor
Computer Engineering Program
School of Engineering
Universidad Distrital Francisco José de Caldas

2026-I



Course Outline

1 Introduction to Complexity Analysis

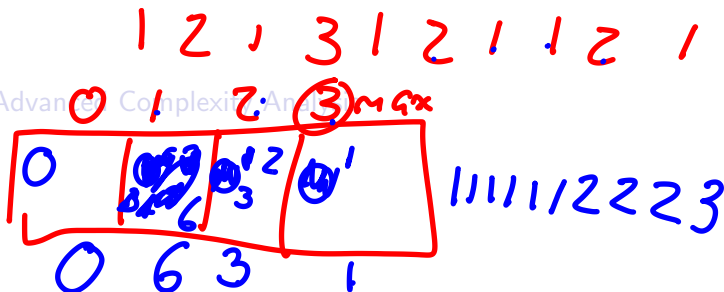
2 Advanced Complexity Analysis



Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis



Why Analyze Algorithms?

The Fundamental Question

Given multiple algorithms that solve the same problem, which one should we choose?

Counting Sort $\rightarrow$ repetitions

① Max $\rightarrow$ numbers: list

②  $\rightarrow$ len = max + 1

③ for ∇ in numbers:
result[len] += 1

④ print result[:len] $\rightarrow$ i.

Why Analyze Algorithms?

The Fundamental Question

Given multiple algorithms that solve the same problem, **which one should we choose?**

- **Performance:** How fast does the algorithm run?
- **Resources:** How much memory does it use?
- **Scalability:** How does it handle large inputs?
- **Predictability:** Can we predict its behavior?

contest

Real World Impact

Condor

- User experience depends on algorithm speed
- Server costs increase with inefficient algorithms
- Mobile apps drain battery with poor algorithms
- Scientific computing requires efficient solutions



Why Analyze Algorithms?

The Fundamental Question

Given multiple algorithms that solve the same problem, **which one should we choose?**

- **Performance:** How **fast** does the algorithm run?
- **Resources:** How much **memory** does it use?
- **Scalability:** How does it handle **large inputs**?
- **Predictability:** Can we predict its **behavior**?

Real-World Impact

- User experience depends on algorithm **speed**
- Server costs increase with inefficient algorithms
- Mobile apps drain battery with poor algorithms
- Scientific computing requires efficient solutions

\$/hour

Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

cost } → machine

Arithmetic: $+$, $-$, $*$, $/$

$=$, $<$, $>$

array[i]

Simple built-in functions



Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

- **Assignment:** $x = 5$
- **Arithmetic:** $+$, $-$, $*$, $/$
- **Comparison:** $==$, $<$, $>$
- **Array Access:** `array[i]`
- **Function Calls:** Simple built-in functions



Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

- **Assignment:** $x = 5$
- **Arithmetic:** $+$, $-$, $*$, $/$
- **Comparison:** $==$, $<$, $>$
- **Array Access:** `array[i]`
- **Function Calls:** Simple built-in functions



Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

- **Assignment:** $x = 5$
- **Arithmetic:** $+$, $-$, $*$, $/$
- **Comparison:** $==$, $<$, $>$ $<=$ $>=$ $!=$
- **Array Access:** `array[i]`
- **Function Calls:** Simple built-in functions



Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

- **Assignment:** $x = 5$
- **Arithmetic:** $+$, $-$, $*$, $/$
- **Comparison:** $==$, $<$, $>$
- **Array Access:** `array[i]`
- **Function Calls:** Simple built-in functions



Elementary Operations Identification I

Definition

An **elementary operation** is a basic computation that takes **constant time** regardless of input size.

Common Elementary Operations:

- **Assignment:** $x = 5$
- **Arithmetic:** $+$, $-$, $*$, $/$
- **Comparison:** $==$, $<$, $>$
- **Array Access:** `array[i]`
- **Function Calls:** Simple built-in functions

1 step

no relevant

sum functions step



Elementary Operations Identification II

Example

Count operations in simple code:

```

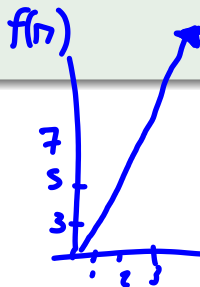
1:  $sum = 0$  // 1 operation
2: for  $i = 0$  to  $n - 1$  do
3:    $sum = sum + array[i]$  // 2 operations per iteration
4: end for
  
```

Handwritten annotations: A red bracket under line 1 points to a circled 'n' with the word 'times' next to it. A red arrow points from the 'do' in line 2 to the same circled 'n'. A pink bracket under lines 3 and 4 points to the same circled 'n'.

Total: 1 + 2n operations

Handwritten annotation: A red circle around '2n' has an arrow pointing to the text 'growth factor'.

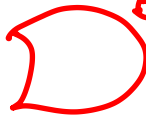
growth factor } loop



Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

- 1: Copy all elements from A to C
- 2: **for** each element ~~in A~~  + 
- 3: Check if element is in C
- 4: **if** element not found **then**
- 5: Add element to C
- 6: **end if**
- 7: **end for**

$$A \cup (B - A)$$



Study Case: Counting Operations in Arrays Union I

Problem

Given two arrays A and B, create array C that contains all elements from both arrays without duplicates.

- 1: Copy all elements from A to C
- 2: **for** each element in B **do**
- 3: Check if element exists in C
- 4: **if** element not found **then**
- 5: Add element to C
- 6: **end if**
- 7: **end for**

$\sim$ each element (n)

$\sim$ n times

Case $A \cap B = \emptyset \rightarrow n + n (2)$

Case $A = B \rightarrow n + n (1)$

Java-loop
Python: `in`



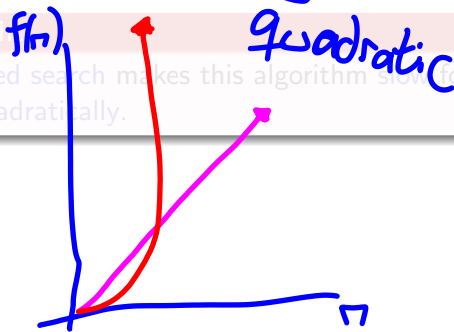
Study Case: Counting Operations in Arrays Union II

Operation Analysis

- Copy from A: n operations *length A*
- For each element in B: m iterations *length B*
- Search in C: Up to n comparisons per element
- Total worst case: $n + m * n = n(1 + m)$ operations

Key Insight

The nested search makes this algorithm slow for large arrays! Performance grows quadratically.



- **Copy from A:** n operations
- **For each element in B:** m iterations
- **Search in C:** Up to n comparisons per element
- **Total worst case:** $n + m * n = n(1 + m)$ operations



The nested search makes this algorithm slow for large arrays. Performance grows quadratically.

$$\left. \begin{array}{l} \text{for } \dots \\ \text{for } \dots \end{array} \right\} \begin{array}{l} n \approx m \\ \approx n \times m^2 \end{array}$$

for for for } 3



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations



Average Case

- Expected performance
- Typical input
- Example: Finding element in middle



Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found

$$I \leq M \leq 1000$$



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

- Expected performance
- Typical input
- Example: Finding element in middle

Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

- Expected performance
- Typical input
- Example: Finding element in middle

Linear

Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found



Best, Average, Worst Case Scenarios

Three Perspectives on Algorithm Performance

Different inputs can lead to dramatically different execution times for the same algorithm.

Best Case

- Minimum operations
- Most favorable input
- Example: Finding element at first position

Average Case

- Expected performance
- Typical input
- Example: Finding element in middle

Worst Case

- Maximum operations
- Least favorable input
- Example: Element not found



Best, Average, Worst Case Scenarios II

Example

Search in array [5, 2, 8, 1, 9]:

- **Best:** Search for 5 $\rightarrow$ 1 comparison
- **Average:** Random element $\rightarrow \sim 2.5$ comparisons
- **Worst:** Search for 7 (not found) $\rightarrow$ 5 comparisons

Linear



Performance Prediction Methods

Approaches to Predict Algorithm Performance

① Theoretical Analysis: Mathematical analysis

② Empirical Analysis: Timing actual executions

③ Simulation: Testing with synthetic data

④ Profiling: Detailed runtime analysis

for $\rightarrow$ 7 times

$x = y * z \rightarrow$ 1 step
 1 step $*$ $=$ 2 steps



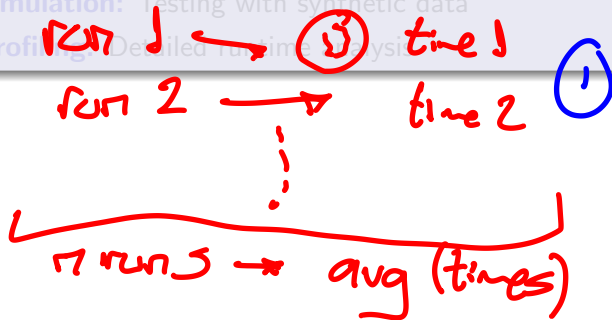
Performance Prediction Methods

Approaches to Predict Algorithm Performance

- ① **Theoretical Analysis:** Mathematical analysis
- ② **Empirical Analysis:** Timing actual executions

③ **Simulation:** Testing with synthetic data

④ **Profiling:** Detailed runtime analysis



Performance Prediction Methods

Approaches to Predict Algorithm Performance

- 1 **Theoretical Analysis:** Mathematical analysis
- 2 **Empirical Analysis:** Timing actual executions
- 3 **Simulation:** Testing with synthetic data
- 4 **Profiling:** Detailed runtime analysis

random
array

fake
data



Performance Prediction Methods

Approaches to Predict Algorithm Performance

- 1 **Theoretical Analysis:** Mathematical analysis
- 2 **Empirical Analysis:** Timing actual executions
- 3 **Simulation:** Testing with synthetic data
- 4 **Profiling:** Detailed runtime analysis

static

run

memory
time
network



Performance Prediction Methods II

Theoretical Analysis

Advantages:

- Machine-independent
- Covers all possible inputs
- Mathematical precision

Limitations:

- Ignores implementation details
- May not reflect real hardware

Empirical Analysis

Advantages:

- Reflects real performance
- Shows actual complexity
- Platform-specific results

Limitations:

- Machine-dependent
- Limited input coverage
- Time-consuming

10 20

7²

best
avg
worst

~1000000
~100.000.000

Best Practice

Use theoretical analysis for algorithm design, then validate with empirical testing.

Performance Prediction Methods II

Theoretical Analysis

Advantages:

- Machine-independent
- Covers all possible inputs
- Mathematical precision

Limitations:

- Ignores implementation details
- May not reflect real hardware

Empirical Analysis

Advantages:

- Reflects real performance
- Shows actual constants
- Platform-specific results

Limitations:

- Machine-dependent
- Limited input coverage
- Time-consuming

Best Practice

Use theoretical analysis for algorithm design, then validate with empirical testing.

Performance Prediction Methods II

Theoretical Analysis

Advantages:

- Machine-independent
- Covers all possible inputs
- Mathematical precision

Limitations:

- Ignores implementation details
- May not reflect real hardware

Empirical Analysis

Advantages:

- Reflects real performance
- Shows actual constants
- Platform-specific results

Limitations:

- Machine-dependent
- Limited input coverage
- Time-consuming

Best Practice

Use theoretical analysis for algorithm design, then validate with empirical testing.

Time vs. Space Trade-offs I

Definition

A time-space trade-off occurs when we can reduce time by using more memory, or reduce memory at the cost of increased time.

Trade-off Decision Factors:

- Available memory constraints
- Time requirements
- Cost of computation vs. storage
- Frequency of algorithm execution

Bank search



Time vs. Space Trade-offs I

Definition

A time-space trade-off occurs when we can reduce time by using more memory, or reduce memory at the cost of increased time.

Trade-off Decision Factors:

- Available memory constraints
- Time requirements
- Cost of computation vs. storage
- Frequency of algorithm execution

Cache



more memory
less time



Time vs. Space Trade-offs II

Example

Fibonacci Calculation:

Space-Efficient (Recursive): ✓

- Uses minimal extra memory
- But recalculates same values repeatedly
- Very slow for large numbers

Time-Efficient (Memoization): ✓

- Stores previously calculated values
- Uses more memory
- Much faster for large numbers



var1 var2 var3



Systematic Complexity Calculation Methods I

Step-by-Step Approach

- ① **Identify:** All loops and recursive calls
- ② **Count:** Operations in each section
- ③ **Multiply:** Operations by execution frequency
- ④ **Sum:** All contributions
- ⑤ **Simplify:** Focus on dominant terms



Nested Loop Analysis:

```

1: for  $i = 0$  to  $n - 1$  do
2:   for  $j = 0$  to  $n - 1$  do
3:      $result = array[i] + array[j]$ 
4:   end for
5: end for

```

- **Outer loop** runs n times
- **Inner loop** runs n times for each outer iteration.
- **Total operations:** $n * n = n^2$ operations



Study Case: Complexity Calculation for QuickSort I

QuickSort Algorithm Structure

- 1: Choose a **pivot** element
- 2: **Partition** array around pivot
- 3: Recursively sort **left** part
- 4: Recursively sort **right** part

$$\begin{aligned}
 & T(n) = T(n-1) + T(1) \\
 & T(n) = T(n-1) + T(n-1) \\
 & T(n) = 2T(n-1)
 \end{aligned}$$

Partition Operation

- Compare each element with pivot
- Rearrange elements (smaller left, larger right)
- This takes n operations for n elements



Study Case: Complexity Calculation for QuickSort I

QuickSort Algorithm Structure

- 1: Choose a **pivot** element
- 2: **Partition** array around pivot
- 3: Recursively sort **left** part
- 4: Recursively sort **right** part



Partition Operation

- Compare each element with pivot *first random*
- Rearrange elements (smaller left, larger right)
- This takes n operations for n elements



Study Case: Complexity Calculation for QuickSort II

 $M - m \rightarrow \text{pivot} \text{ (e)}$

Best Case

- Pivot divides array in half
- Balanced partitions
- Fast performance



Worst Case

- Pivot is always smallest/largest
- Unbalanced partitions
- Slow performance



Outline

1 Introduction to Complexity Analysis

2 Advanced Complexity Analysis



Big O Notation: Definition and Meaning

Definition

Big O notation describes the upper bound of an algorithm's growth rate for large inputs.

What Big O Tells Us:

- How algorithm performance scales with input size
- Worst-case growth rate
- Ignores constant factors and smaller terms
- Focuses on dominant behavior for large inputs



Big O Notation: Definition and Meaning

Definition

Big O notation describes the upper bound of an algorithm's **growth rate** for large inputs.

What Big O Tells Us:

- How algorithm performance scales with input size
- Worst-case growth rate
- Ignores constant factors and smaller terms
- Focuses on dominant behavior for large inputs

$$\boxed{\begin{matrix} x^2 + x + 2 \rightarrow x^2 \\ x^3 + 2x^2 + 3x + 1 \rightarrow x^3 \end{matrix}}$$



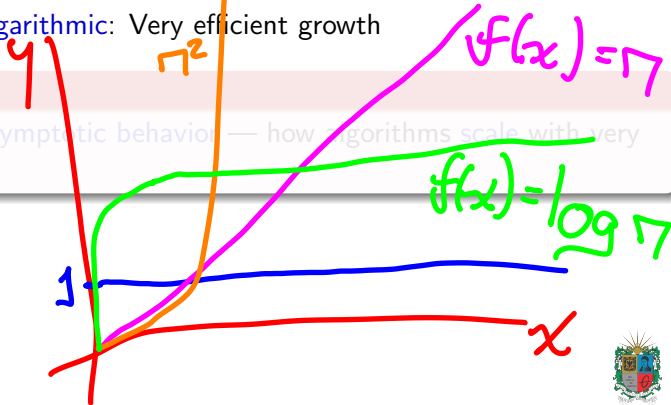
Big O Notation: Classifications

Common Big O Classifications:

- $O(1)$ — **Constant**: Always same time
- $O(n)$ — **Linear**: Time grows with input size
- $O(n^2)$ — **Quadratic**: Time grows with square of input
- $O(\log n)$ — **Logarithmic**: Very efficient growth

Key Insight

Big O focuses on asymptotic behavior — how algorithms scale with very large inputs.



Big O Notation: Classifications

Common Big O Classifications:

- $O(1)$ — **Constant**: Always same time
- $O(n)$ — **Linear**: Time grows with input size
- $O(n^2)$ — **Quadratic**: Time grows with square of input
- $O(\log n)$ — **Logarithmic**: Very efficient growth

$n \cdot \log n$ ✓

Key Insight

Big O focuses on asymptotic behavior — how algorithms scale with very large inputs.



Upper Bound Concept and Mathematical Definition

Understanding Upper Bounds

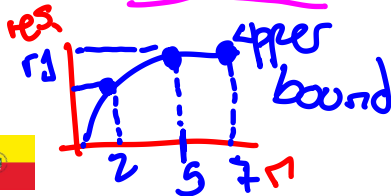
An upper bound guarantees that an algorithm will never perform worse than the bound.

Big O Meaning:

- Sets a ceiling on performance ✓
- Guarantees maximum time
- Used for worst-case analysis
- Helps compare algorithms

Common Growth Orders:

- $O(1)$ — Constant
- $O(\log n)$ — Logarithmic
- $O(n)$ — Linear
- $O(n \log n)$ — Linearithmic
- $O(n^2)$ — Quadratic
- $O(2^n)$ — Exponential



Study Case: First Letter in a Sentence Complexity

Problem

Find the first alphabetical letter in a given sentence string.

```

1: for each character in sentence do
2:   if character is a letter then
3:     return character
4:   end if
5: end for
6: return null
  
```

"hello"

↓ → 7 = len(s)

break

not found

Best: First character is a letter

- Only check 1 character
- Constant time: $O(1)$

Worst: No letters in sentence

- Check every character
- Linear time: $O(n)$

Even though best case is $O(1)$, we report $O(n)$ because Big O represents the worst-case upper bound.

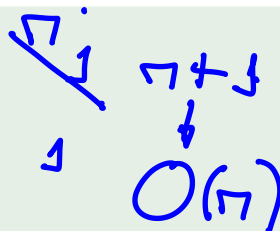
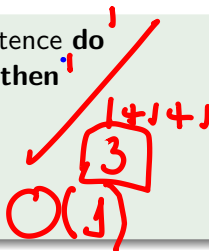
Study Case: First Letter in a Sentence Complexity

Problem

Find the **first alphabetical letter** in a given sentence string.

```

1: for each character in sentence do
2:   if character is a letter then
3:     return character
4:   end if
5: end for
6: return null
  
```



Best: First character is a letter

- Only check 1 character
- Constant time: $O(1)$

Worst: No letters in sentence

- Check every character
- Linear time: $O(n)$

upper bound

Even though best case is $O(1)$, we report $O(n)$ because $O(n)$ represents the worst-case upper bound.

Study Case: First Letter in a Sentence Complexity

Problem

Find the **first alphabetical letter** in a given sentence string.

```

1: for each character in sentence do
2:   if character is a letter then
3:     return character
4:   end if
5: end for
6: return null
  
```

Best: First character is a letter

- Only check **1 character**
- Constant time: $O(1)$

Worst: No letters in sentence

- Check **every character**
- Linear time: $O(n)$

Even though best case is $O(1)$, we report $O(n)$ because Big O represents the **worst-case upper bound**.

Study Case: Linear Search Algorithm Complexity

Problem

Search for a **target element** in an **unsorted** array.

```

1: for each element in array do
2:   if element equals target then
3:     return index
4:   end if
5: end for
6: return -1
  
```

- **Best Case:** Element at first position, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element not in array, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in middle, $n/2$ comparisons $\rightarrow O(n)$



Study Case: Linear Search Algorithm Complexity

Problem

Search for a **target element** in an **unsorted** array.

```

1: for each element in array do
2:   if element equals target then
3:     return index
4:   end if
5: end for
6: return -1
  
```

$1 + 1 + 4 \downarrow$
 $3 \Rightarrow O(1)$

$n \downarrow$
 $n + 1 \downarrow$
 $O(n)$

$n/2 \downarrow$
 $n/2 + 1 \downarrow$
 $n/2 + 2 \downarrow$

- **Best Case:** Element at **first position**, 1 comparison $\rightarrow O(1)$
- **Worst Case:** Element **not in array**, n comparisons $\rightarrow O(n)$
- **Average Case:** Element in **middle**, $n/2$ comparisons $\rightarrow O(n)$

$O(n)$



Study Case: Binary Search Algorithm Complexity I

Problem

Search for a target element in a **sorted** array.

```

1: while search space not empty do
2:   Find middle element
3:   if middle equals target then
4:     return index
5:   else if middle > target then
6:     search right half
7:   else
8:     search left half
9:   end if
10: end while
11: return -1
  
```

$$\lceil \log_2(n) \rceil$$

$$O(\log n)$$

$$\lceil \log_2(n) \rceil$$

$$\lceil \log_2(n) \rceil$$

$$\lceil n \rceil = \log_e$$

$$a^b = c \Rightarrow \log_a c = b$$



Study Case: Binary Search Algorithm Complexity II

Each step eliminates half of the remaining elements:

- Step 1: n elements
- Step 2: $n/2$ elements
- Step 3: $n/4$ elements
- Step k : $n/2^k$ elements

Algorithm stops when we reach 1 element.

$$\begin{array}{l|l}
 \log n & \log n \\
 2 \log n/2 & \log n/2 \\
 4 \log n/4 & \log n/4 \\
 \vdots & \vdots
 \end{array}$$



Big Omega (Ω) and Big Theta (Θ) Notations I

Big Omega (Ω)

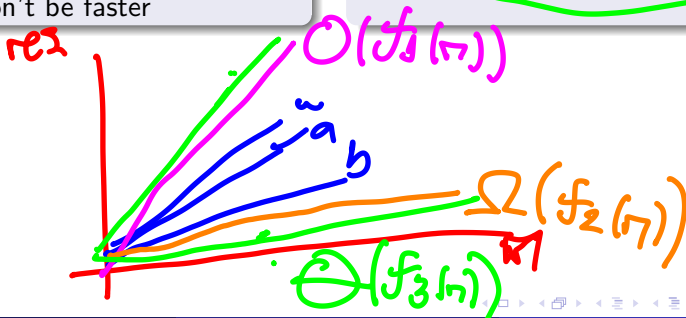
Lower Bound:

- Minimum time required
- Best-case analysis
- "At least this fast"
- Guarantees algorithm won't be faster

Big Theta (Θ)

Tight Bound:

- Exact growth rate
- Best and worst are same
- "Exactly this fast"
- Most precise description



Big Omega (Ω) and Big Theta (Θ) Notations II

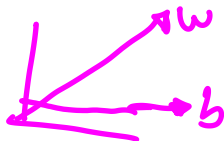
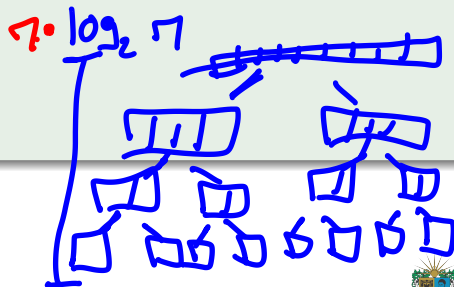
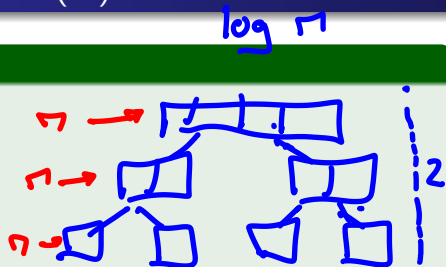
Example

Merge Sort:

- Best case: $\Theta(n \log n)$
- Worst case: $O(n \log n)$
- Since best = worst: $\Theta(n \log n)$

Linear Search:

- Best case: $\Omega(1)$
- Worst case: $O(n)$
- No tight bound (best $\neq$ worst)



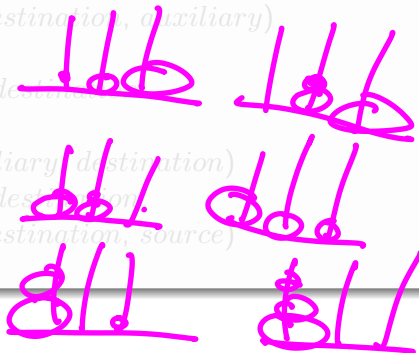
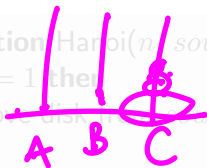
Study Case: Tower of Hanoi Problem I

Problem

Move n disks from source rod to destination rod using auxiliary rod. Rules: only move one disk at a time, never place larger disk on smaller one.

```

1: function Hanoi( $n$ , source, destination, auxiliary)
2: if  $n = 1$  then
3:   move disk from source to destination
4: else
5:   Hanoi( $n - 1$ , source, auxiliary, destination)
6:   move disk from source to destination
7:   Hanoi( $n - 1$ , auxiliary, destination, source)
8: end if
  
```



Study Case: Tower of Hanoi Problem I

Problem

Move n disks from *source* rod to *destination* rod using *auxiliary* rod. Rules: only move *one disk* at a time, never place *larger disk* on smaller one.

```
1: function Hanoi( $n$ , source, destination, auxiliary)
2: if  $n = 1$  then
3:   move disk from source to destination
4: else
5:   Hanoi( $n - 1$ , source, auxiliary, destination)
6:   move disk from source to destination
7:   Hanoi( $n - 1$ , auxiliary, destination, source)
8: end if
```



Study Case: Tower of Hanoi Problem II

Recurrence Analysis

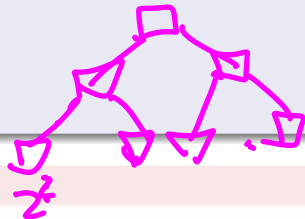
- $T(1) = 1$ (base case: move one disk)
- $T(n) = 2T(n-1) + 1$ (two recursive calls + one move)

Solution:

- $T(n) = 2^n - 1$
- Complexity: $O(2^n)$ — exponential!

Real-World Implication

For 64 disks: $2^{64} - 1 \approx 18$ billion billion moves!



Study Case: Tower of Hanoi Problem II

Recurrence Analysis

- $T(1) = 1$ (base case: move one disk)
- $T(n) = 2T(n - 1) + 1$ (two recursive calls + one move)

Solution:

- $T(n) = 2^n - 1$
- Complexity: $O(2^n)$ — exponential!

Real-World Implication

For 64 disks: $2^{64} - 1 \approx 18$ billion ~~million~~ moves!



Solving Recurrence Relations

Common Solution Methods

- 1 **Substitution Method:** Guess and verify
- 2 **Recursion Tree:** Visualize recursive calls
- 3 **Master Theorem:** Direct formula for common cases
- 4 **Characteristic Equation:** For homogeneous recurrences

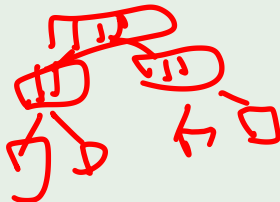


Solving Recurrence Relations

Example

Recursion Tree Method for $T(n) = 2T(n/2) + n$:

- Level 0: n work
- Level 1: $2 \times \cancel{n/2} = n$ work
- Level 2: $4 \times \cancel{n/4} = n$ work
- ...
- Total levels: $\log n$
- Total work: $n \times \log n = O(n \log n)$



Method Selection

- Simple cases: Direct solving
- Divide-and-conquer: Master theorem
- Complex cases: Substitution method

Solving Recurrence Relations

Example

Recursion Tree Method for $T(n) = 2T(n/2) + n$:

- Level 0: n work
- Level 1: $2 \times (n/2) = n$ work
- Level 2: $4 \times (n/4) = n$ work
- ...
- Total levels: $\log n$
- Total work: $n \times \log n = O(n \log n)$

Method Selection

- Simple cases: Direct solving
- Divide-and-conquer: Master theorem
- Complex cases: Substitution method

Master Theorem Introduction I

Master Theorem

For recurrences of the form $T(n) = aT(n/b) + f(n)$:

Case 1: $f(n)$ grows **slower** than $n^{(\log_b a)}$

- $T(n) = O(n^{(\log_b a)})$

Case 2: $f(n)$ grows at **same rate**

- $T(n) = O(n^{(\log_b a)} \log n)$

Case 3: $f(n)$ grows **faster**

- $T(n) = O(f(n))$



Master Theorem Introduction II

Examples

Merge Sort:

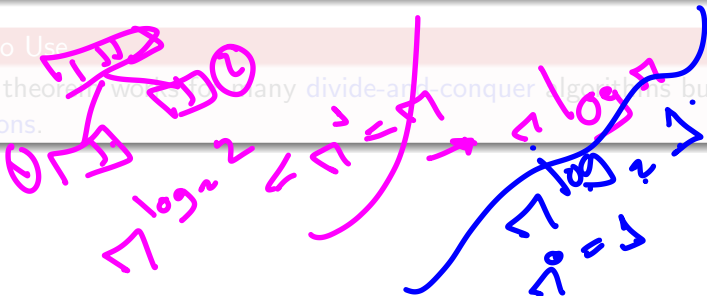
- $T(n) = 2T(n/2) + n$
- $a = 2, b = 2, f(n) = n$
- Case 2: $O(n \log n)$

Binary Search:

- $T(n) = T(n/2) + 1$
- $a = 1, b = 2, f(n) = 1$
- Case 2: $O(\log n)$

When to Use

Master theorem works for many divide-and-conquer algorithms but has limitations.



Master Theorem Introduction II

Examples

Merge Sort:

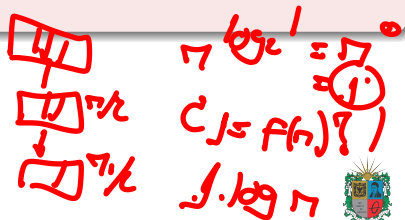
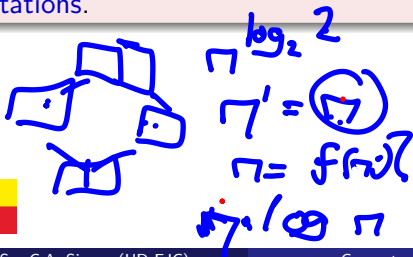
- $T(n) = 2T(n/2) + n$
- $a = 2, b = 2, f(n) = n$
- Case 2: $O(n \log n)$

Binary Search:

- $T(n) = T(n/2) + 1$
- $a = 1, b = 2, f(n) = 1$
- Case 2: $O(\log n)$

When to Use

Master theorem works for many divide-and-conquer algorithms but has limitations.



Recursive Algorithm Complexity Calculation I

Systematic Approach

- ① **Identify:** Base case(s) and recursive case(s)
- ② **Set up:** Recurrence relation
- ③ **Solve:** Using appropriate method
- ④ **Verify:** Check solution makes sense

Expansion

Master Theorem

$$f(n-1) + f(n-2)$$

$$\times 2T(n-1) + 1$$

$$T(n) = T(n-1) + T(n-2) \\ T(n-1) + T(n-2) \cdot T(n-3) + T(n-4) \\ \vdots$$

$$\nearrow n^n$$



Recursive Algorithm Complexity Calculation II

Merge Sort Analysis

Step 1: Identify cases

- **Base:** Array size $1 \rightarrow O(1)$
- **Recursive:** Divide, sort halves, merge

Step 2: Set up recurrence

- $T(n) = 2T(n/2) + O(n)$

Step 3: Solve using Master Theorem

- Case 2: $T(n) = O(n \log n)$

Step 4: Verify

- Makes sense: **logarithmic levels**, **linear work** per level



Summary I

Key Takeaways

- **Complexity Analysis:** Essential for algorithm **comparison** and **selection**
- **Big O Notation:** Describes **growth rate** for large inputs
- **Recursion:** Powerful technique requiring careful **base case** design
- **Recurrence Relations:** Mathematical tools for **recursive algorithm** analysis



Summary II

Practical Skills Gained

- Count operations systematically
- Classify algorithms by growth rate
- Design recursive solutions correctly
- Analyze recursive algorithm complexity
- Apply Master Theorem to common patterns

Next Steps

- Practice complexity analysis on various algorithms
- Implement recursive solutions with proper optimization
- Study advanced recurrence solving techniques
- Apply knowledge to real-world algorithm selection



Summary II

Practical Skills Gained

- Count operations systematically
- Classify algorithms by growth rate
- Design recursive solutions correctly
- Analyze recursive algorithm complexity
- Apply Master Theorem to common patterns

Next Steps

- Practice complexity analysis on various algorithms
- Implement recursive solutions with proper optimization
- Study advanced recurrence solving techniques
- Apply knowledge to real-world algorithm selection



Outline

- 1 Introduction to Complexity Analysis
- 2 Advanced Complexity Analysis



Thanks!

Questions?



Repo: <https://github.com/EngAndres/ud-public/tree/main/courses/computer-sciences-i>

